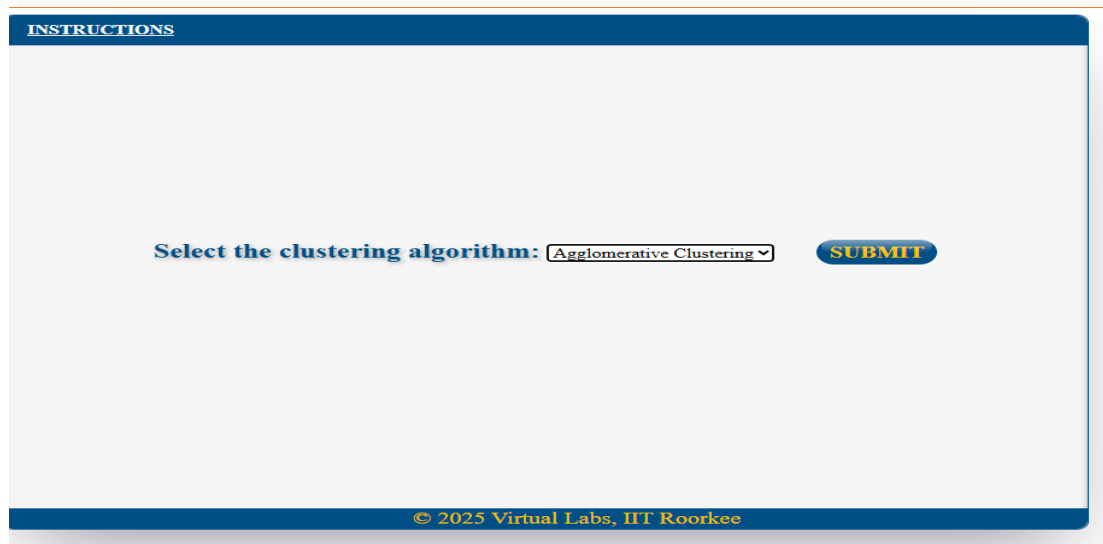


Ex.No: 12

OBSERVATION:(Simulation)



INSTRUCTIONS

Select the clustering algorithm: **SUBMIT**

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Fig 1: Selecting the clustering algorithm (Agglomerative Clustering)



CHOOSE  COMPUTE  PLOT  INSTRUCTIONS

Select the linkage type: **SUBMIT**

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Fig 2: Selecting the Linkage type (Single Linkage)

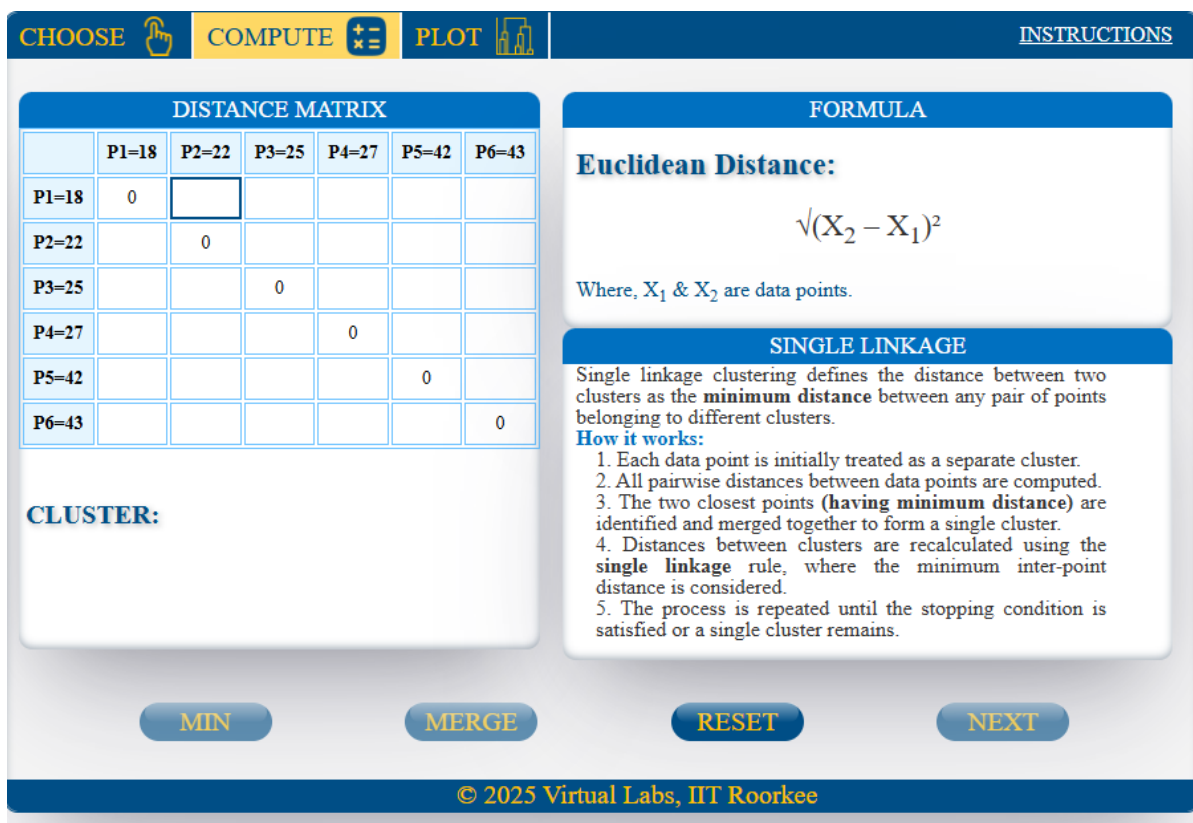


Fig 3: Initial distance matrix and formula display

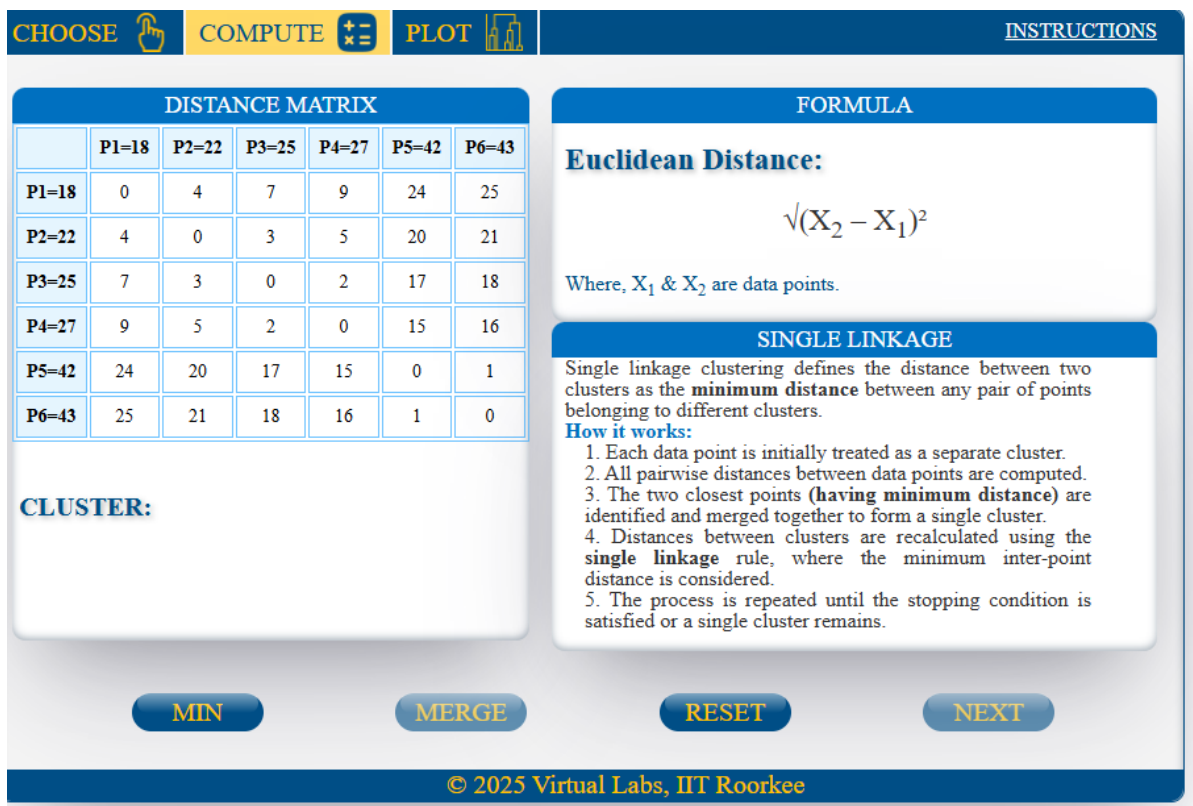


Fig 4: Calculated distance matrix for all data points

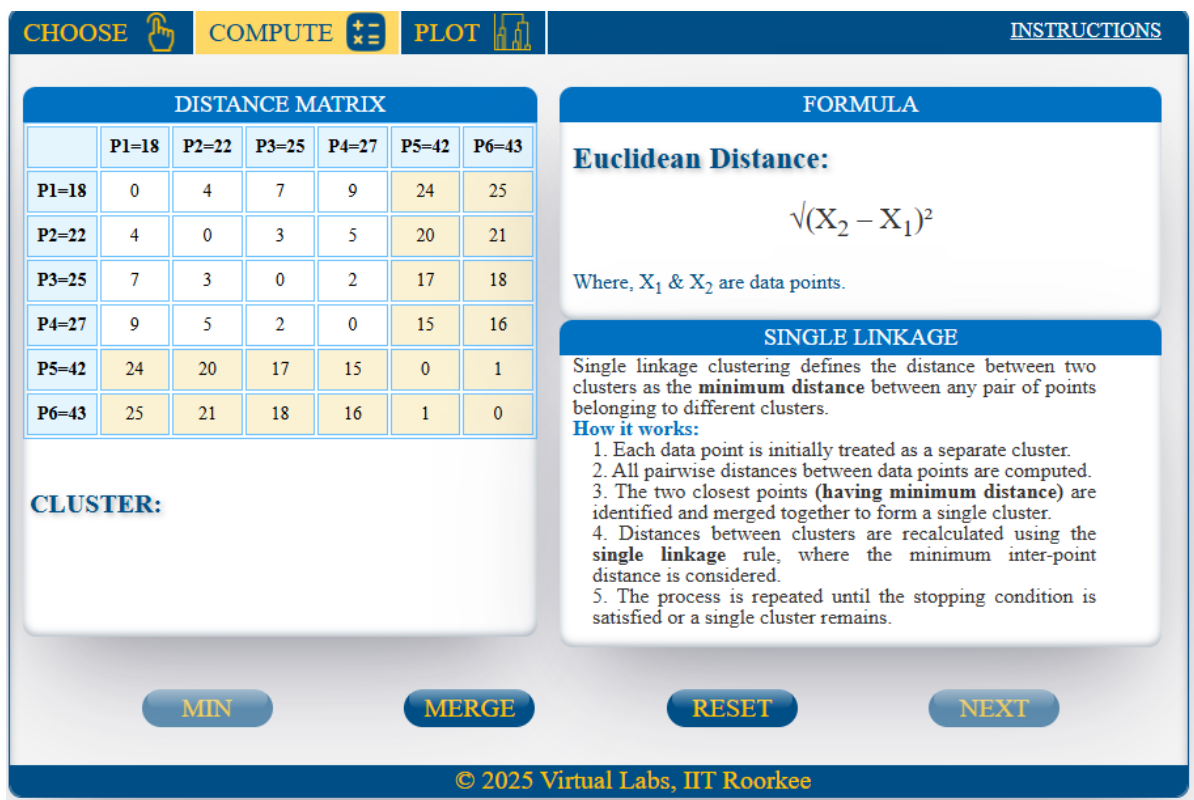


Fig 5: Finding the minimum distance in the distance matrix

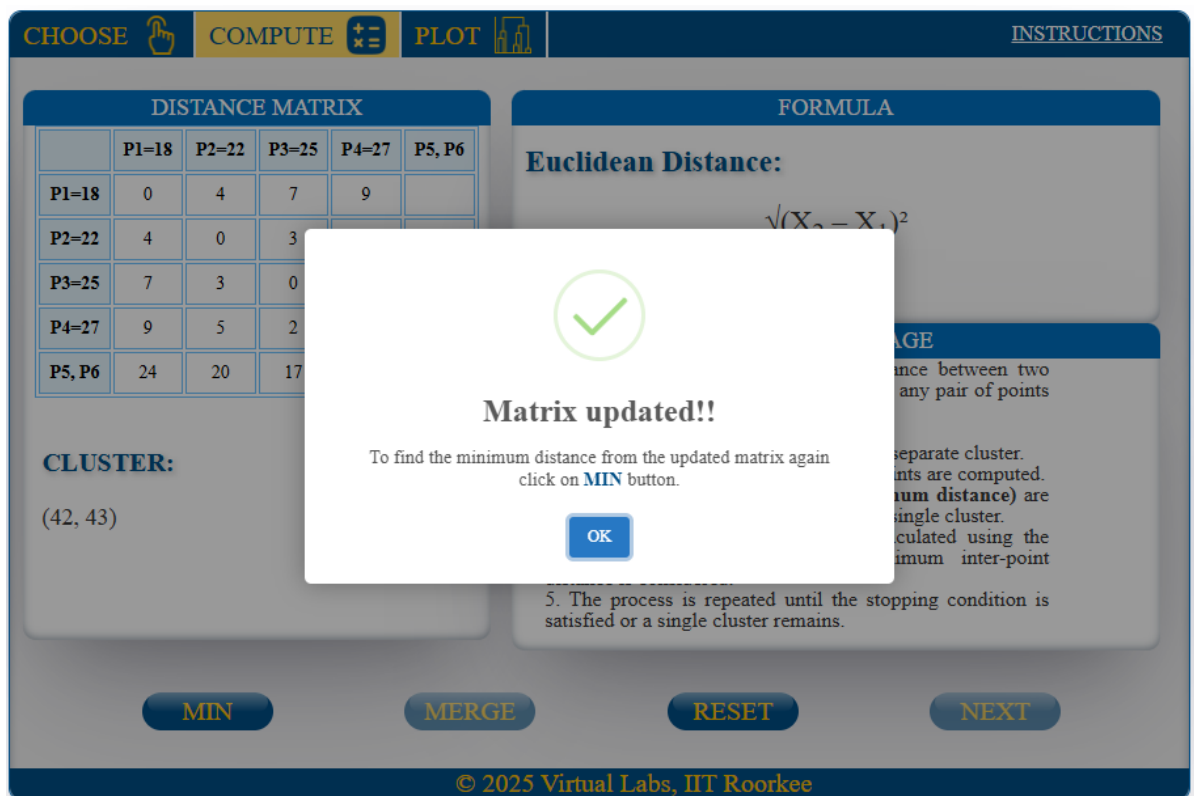


Fig 6: Popup confirmation after updating the distance matrix for cluster(42, 43)

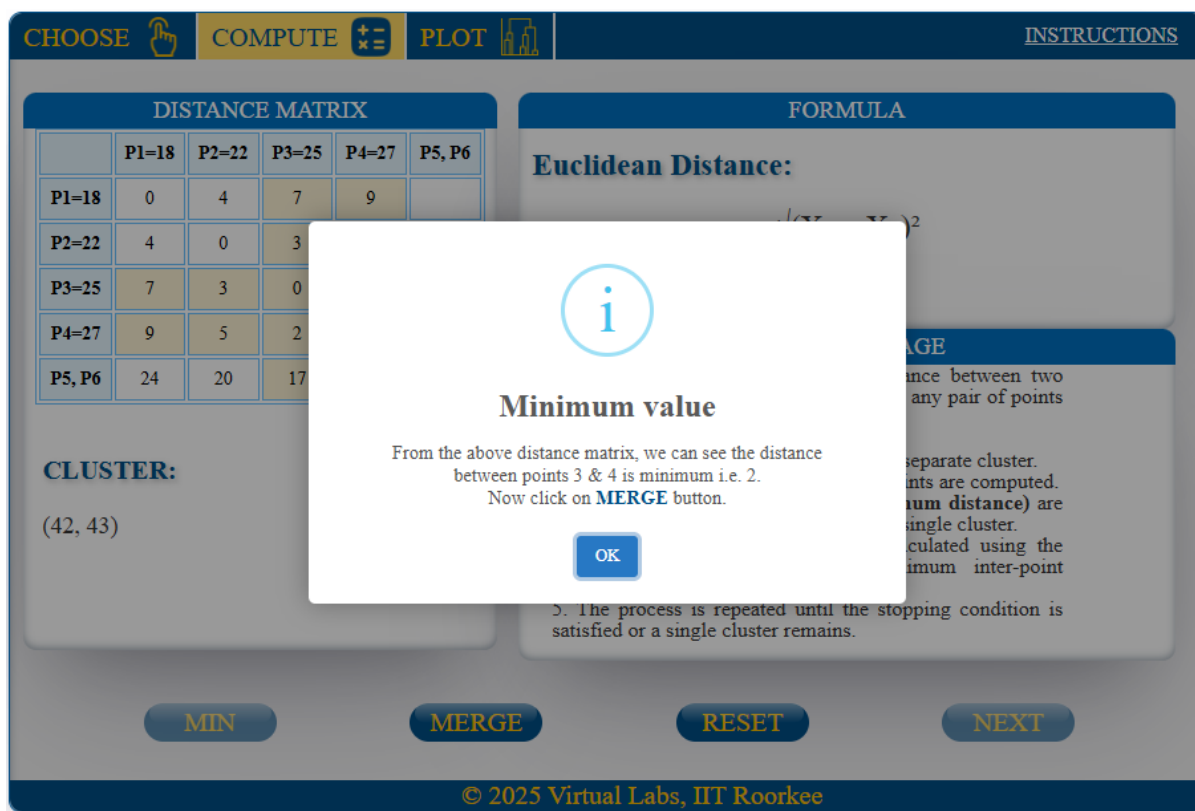


Fig 7: Highlighting minimum distance between points 3 and 4

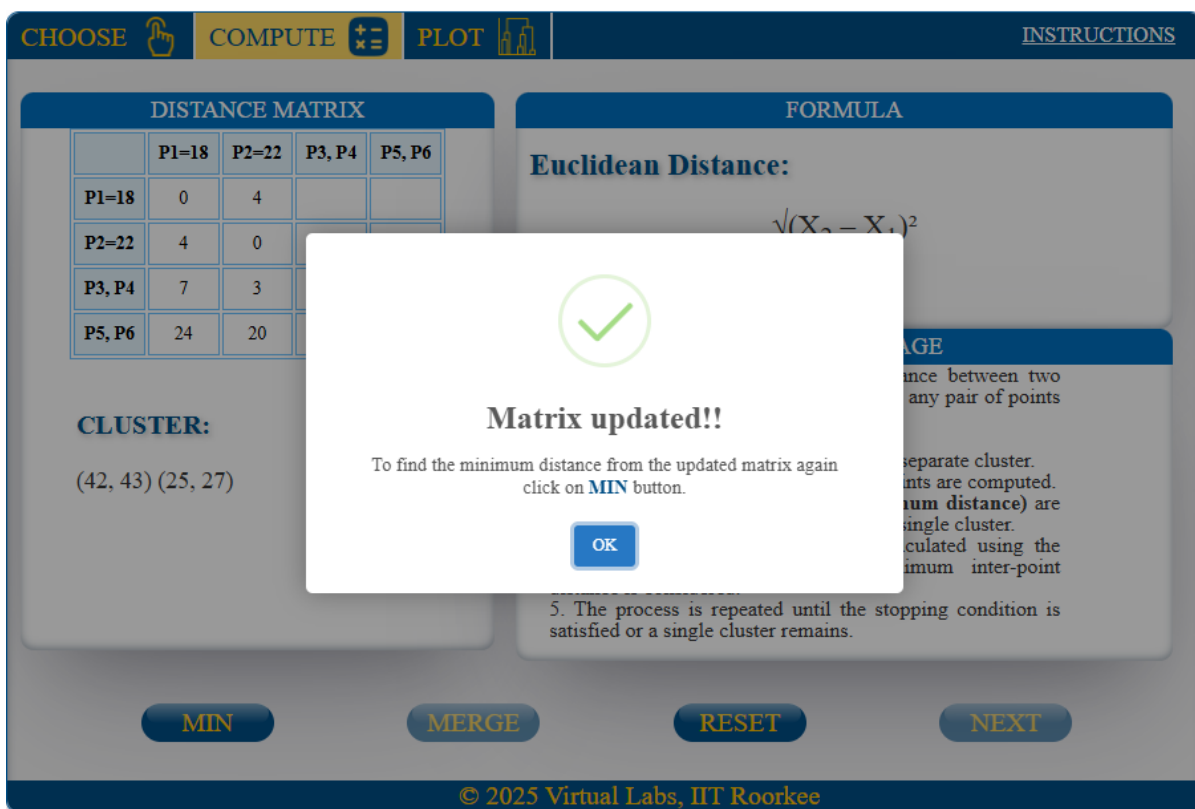


Fig 8: Popup confirmation after updating matrix for cluster (25, 27)

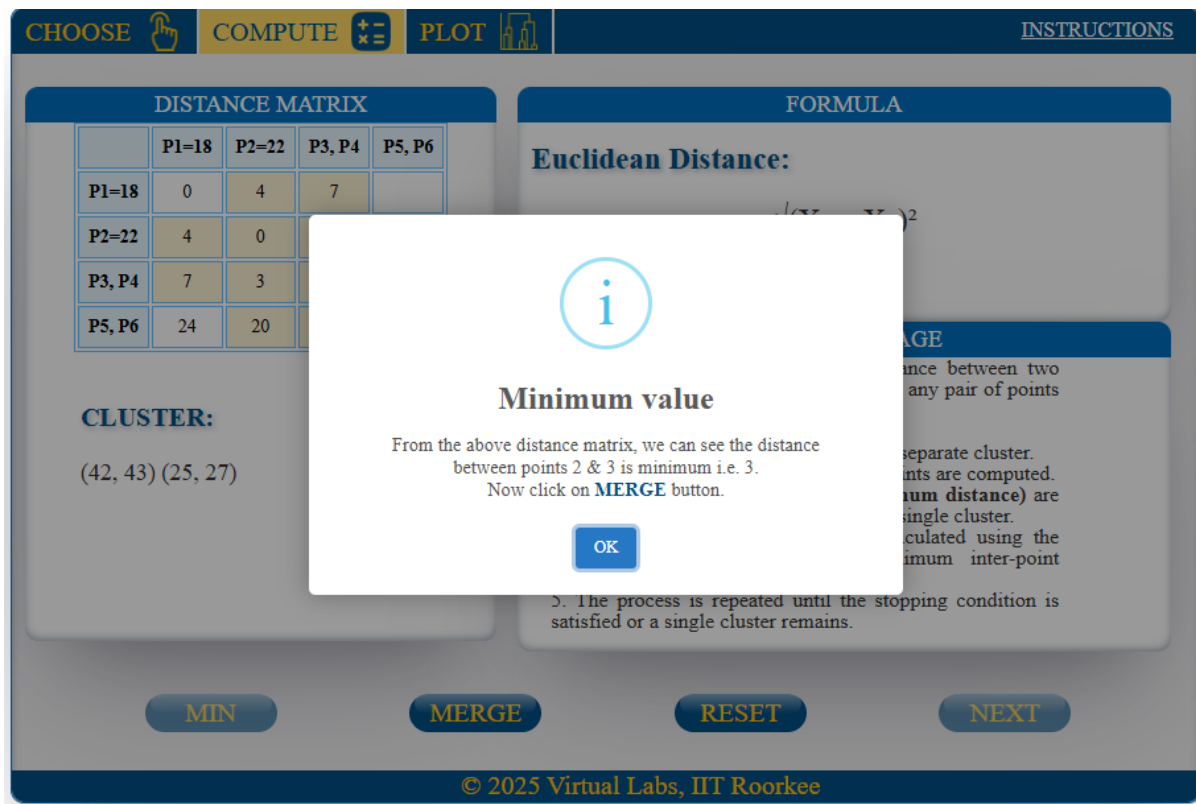


Fig 9: Highlighting minimum distance between point 2 and cluster (25, 27)

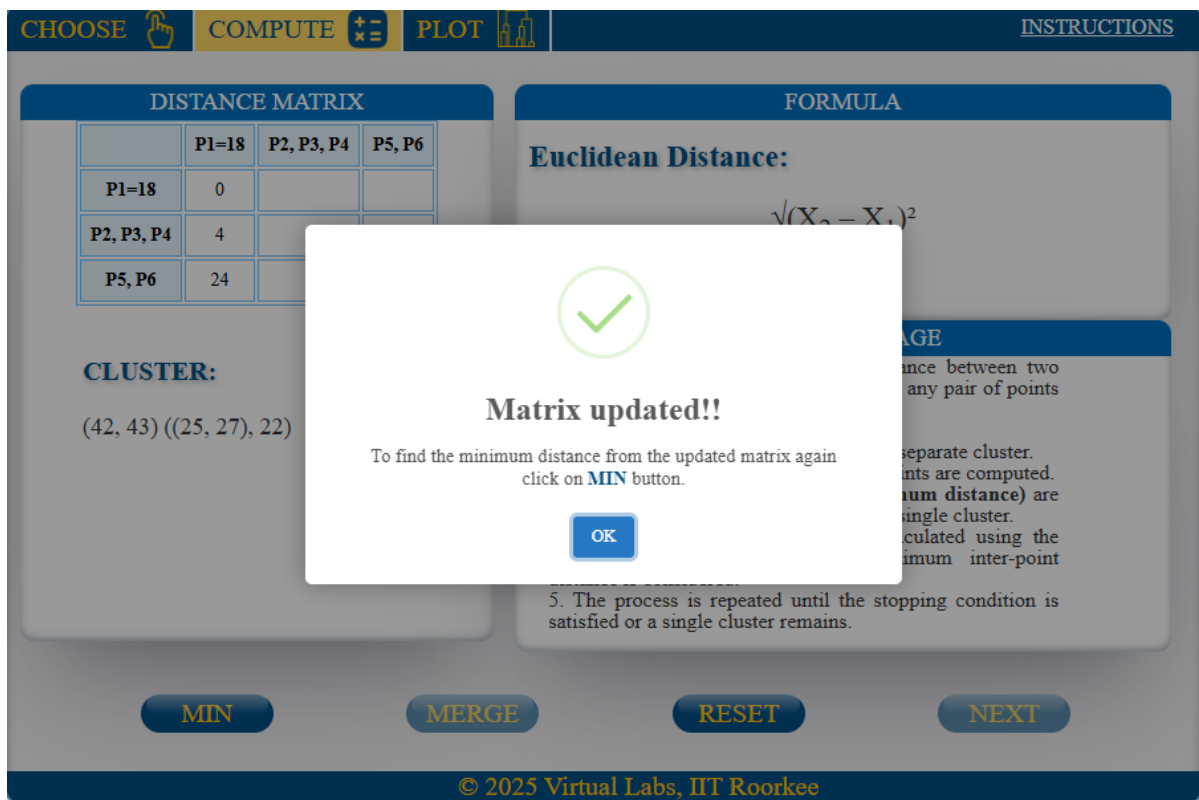


Fig 10: Popup confirmation after merging cluster ((25, 27), 22)

CHOOSE

COMPUTE

PLOT

INSTRUCTIONS

DISTANCE MATRIX

	P1=18	P2, P3, P4	P5, P6
P1=18	0	4	24
P2, P3, P4	4		
P5, P6	24		

CLUSTER:

(42, 43) ((25, 27), 22)

FORMULA

Euclidean Distance:

$$\sqrt{(X_2 - X_1)^2 + (Y_2 - Y_1)^2}$$

MERGE

Distance between two points is calculated using the minimum inter-point distance between any pair of points.

separate cluster. Points are computed. (minimum distance) are single cluster. calculated using the minimum inter-point

5. The process is repeated until the stopping condition is satisfied or a single cluster remains.

MIN

MERGE

RESET

NEXT

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Minimum value

From the above distance matrix, we can see the distance between points 1 & 2 is minimum i.e. 4. Now click on **MERGE** button.

OK

Fig 11: Highlighting minimum distance between point 1 and cluster((25, 27),22)

CHOOSE

COMPUTE

PLOT

INSTRUCTIONS

DISTANCE MATRIX

	P1, P2, P3, P4	P5, P6
P1, P2, P3, P4		15
P5, P6		

CLUSTER:

(42, 43) (((25, 27), 22), 18)

FORMULA

Euclidean Distance:

$$\sqrt{(X_2 - X_1)^2 + (Y_2 - Y_1)^2}$$

MERGE

Distance between two points is calculated using the minimum inter-point distance between any pair of points.

separate cluster. Points are computed. (minimum distance) are single cluster. calculated using the minimum inter-point

5. The process is repeated until the stopping condition is satisfied or a single cluster remains.

MIN

MERGE

RESET

NEXT

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Matrix updated!!

To find the minimum distance from the updated matrix again click on **MIN** button.

OK

Fig 12: Popup confirmation after updating matrix for cluster(((25, 27), 22), 18)

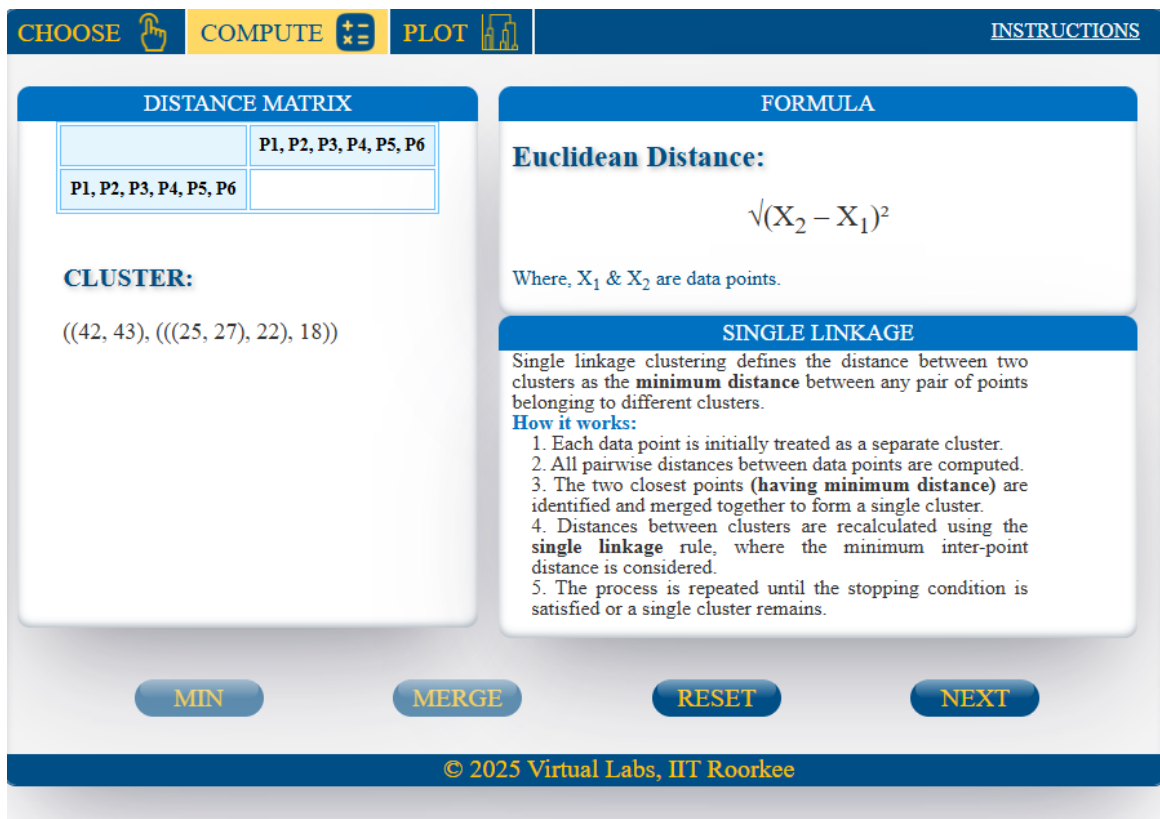


Fig 13: Final distance matrix calculation before total cluster merge



Fig 14: Prompt message asking to plot the dendrogram



Fig 15: Viewing selected data points List

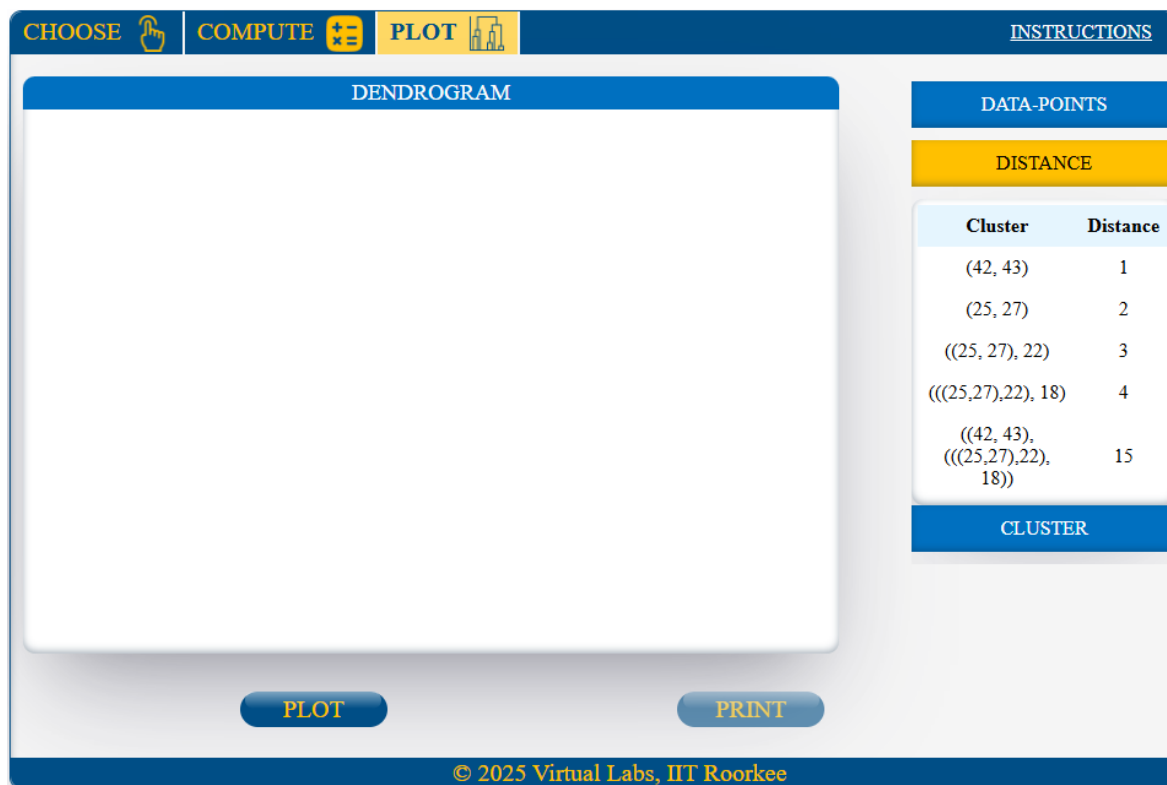


Fig 16: Viewing cluster distance values table



Fig 17: Viewing final merged cluster output string

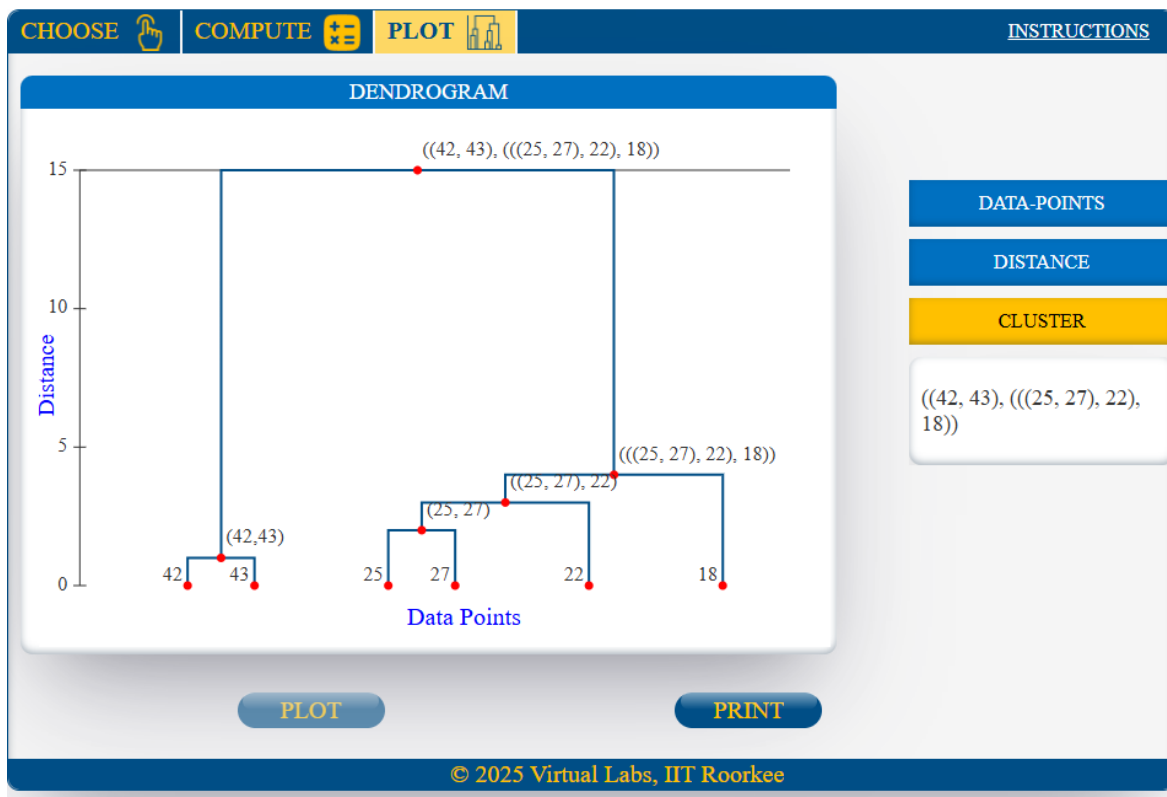


Fig 18: Final plotted dendrogram showing full hierarchical cluster tree

Simulation link: <https://ml-iitr.vlabs.ac.in/exp/hierarchical-clustering/simulation.html>